Classification Methods: k-NN, Logistic, Discriminants

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Classification

- We will talk about different classification strategies: k nearest neighbours, logistic regression, discriminant analysis.
- Before we start k-nn, we will do a quick review of a related statistical concept called the kernel density estimation or KDE.
- We'll also do a quick review of maximum likelihood estimation.

A little detour: Kernel density estimation

Density Estimation

- Suppose a random sample $X_1, \ldots, X_n$ are from a population with an unknown continuous probability density function $f(x)$.
- You want to estimate $f(x)$.

Histogram

- The most basic idea is drawing a histogram.
- Partition the range of data into sub-intervals $a_1 < a_2 < \ldots < a_k$.
- For $x \in (a_i, a_{i+1}]$,

$$\hat{f}(x) = \frac{\text{No. of observations} \in (a_i, a_{i+1}]}{n(a_{i+1} - a_i)}$$

Histogram

- Typically, the intervals have equal length $(a_{i+1} - a_i) = 2h$ for $i = 1, \ldots, k$. The equal interval length h is often referred to as bandwidth. In this case, for any x within the data range,

$$\hat{f}(x) = \frac{\text{No. of observations within } h \text{ of } x}{n \times 2h}$$
$$= \frac{1}{2nh} \sum_{i=1}^{n} I(|X_i - x| \leq h).$$

- Narrow intervals (smaller h): histogram has a choppy appearance, large variance.
- Large intervals (larger h): histogram may lose the local feature, large bias.
- Generally, a larger sample size n requires a smaller h, thus more intervals.
- Suggested $h = \frac{1.75}{n^{1/3}} S$, S is the sample standard deviation.

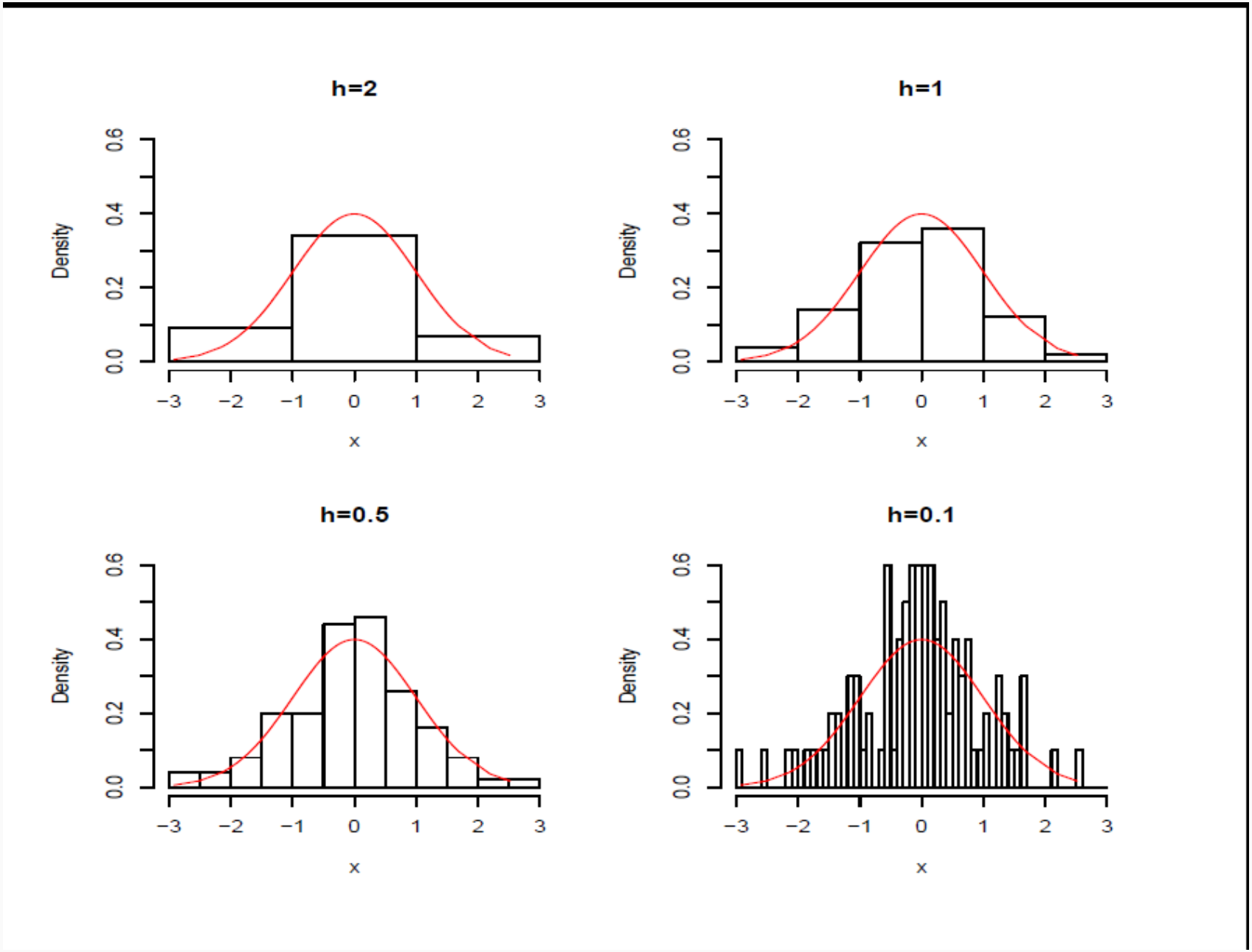


Figure 1

Kernel Density Estimation

- Drawback of histogram: density estimation is piecewise constant and thus unsmooth.
- Main idea of kernel density estimation: instead of counting number of observations within h of x, we take a certain weighted average of data points near x to estimate $f(x)$.
- Usually we assign higher weights to data points closer to x, and lower weights to those further away from x.

Kernel Density Estimation

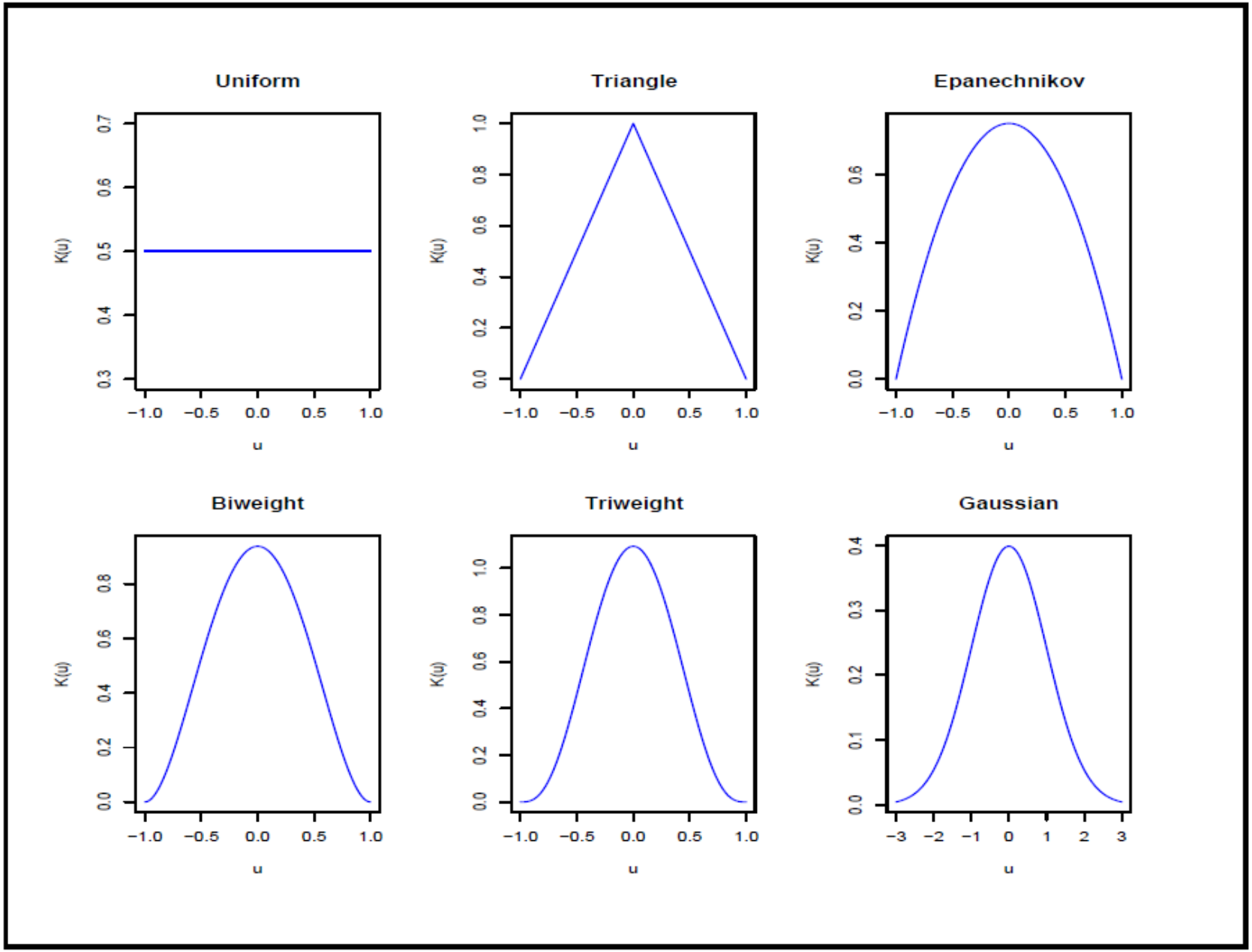
- Kernel Density Estimate:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K(\frac{X_i - x}{h}).$$

  where $K(\cdot)$ is a Kernel function.

- Choices of kernel function
  1. Uniform: $K(u) = 1/2I(u \leq 1)$: reduces to histogram.
  2. Gaussian: $K(u) = \phi(u)$ (standard normal)
  3. Triangle: $K(u) = (1 - |u|)I(|u| \leq 1)$.
  4. Epanechnikov: $K(u) = 3/4(1 - u^2)I(|u| \leq 1)$.
  5. Biweight: $K(u) = 15/16(1 - u^2)^2 I(|u| \leq 1)$.
  6. Triweight: $K(u) = 35/32(1 - u^2)^3 I(|u| \leq 1)$.

Kernels


Uniform
Triangle
Epanechnikov
Biweight
Triweight
Gaussian
K(u)
u

Bandwidth selection

- Choice of h:
- A rule of thumb (Harfle, 1991):

$$h = \frac{1.05}{n^{1/5}} S,$$

where S is the same standard deviation of the sample, or can also be taken as more robust version of standard deviation, e.g.

$$\text{Scaled IQR} = (X_{(0.75)} - X_{(0:25)})/1.34 \tag{1}$$

$$\text{Scaled MAD} = 1.4826 \times \text{Med}\{|X_i - \text{Med}(X_i)|\} \tag{2}$$

MAD = median absolute deviation. IQR = Inter-quartile range.

Look at R examples

https://dattahub.github.io/data-analytics/kde_demo.html

Classification

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- Examples:
  1. Medical Diagnosis: Classify which condition.
  2. Online banking: Determine whether a transaction is fraudulent.
  3. Web search: Based on a user's history, location, and the string of a web search, predict which link a person is likely to click.
  4. Online advertising: Predict whether a user will click on an ad or not.

Classification

- The output y is categorical - takes values in a discrete set.
- Example: Predict the brand of a product, Cancer genetics - predict cancer or normal (binary), Physics - Higgs Boson or other particles, etc.
- The usual model $y = f(x) + \epsilon$ is less relevant as Y is categorical. $Y \in \{1, 2, \ldots, k\}$.
- Instead model $P(Y = i \mid X)$ Conditional distribution of Y given X $i = 1, \ldots, k$.
- $\hat{y}_i$: predicted label for x_i.

Loss function

- Squared error: $(y_i - \hat{f}(x_i))^2$
- Many ways of measuring prediction error when your predicted values are class labels.
- Most common way: $0-1$ loss

$$\mathbb{E}(\mathbf{1}(y_0 \neq \hat{y}_0))$$

- As with squared error, we can compute average test prediction error (called test error rate under 0-1 loss) using previously unseen test data:

$$\frac{1}{m}\sum_{i=1}^{m}\mathbf{1}(y_i^{'} \neq \hat{y}_i^{'}) \tag{3}$$

- Similarly, we can compute the (usually optimistic) training error rate:

$$\frac{1}{n}\sum_{i=1}^{n}\mathbf{1}(y_i \neq \hat{y}_i)$$

- A good classifier is one for which the test error (3) is smallest.

The Bayes Classifier

- It is possible to show that the test error rate in (3) is minimized, on average, by a very simple classifier that assigns each observation to the most likely class, given its predictor values.
- The Bayes classifier predicts

$$\hat{y}_0 = \underset{j}{\operatorname{argmax}}\, P(Y = j \mid X = x_0)$$

- In practice we don't know $P(Y \mid X)$, but many classification methods operate by estimating these conditional probabilities.

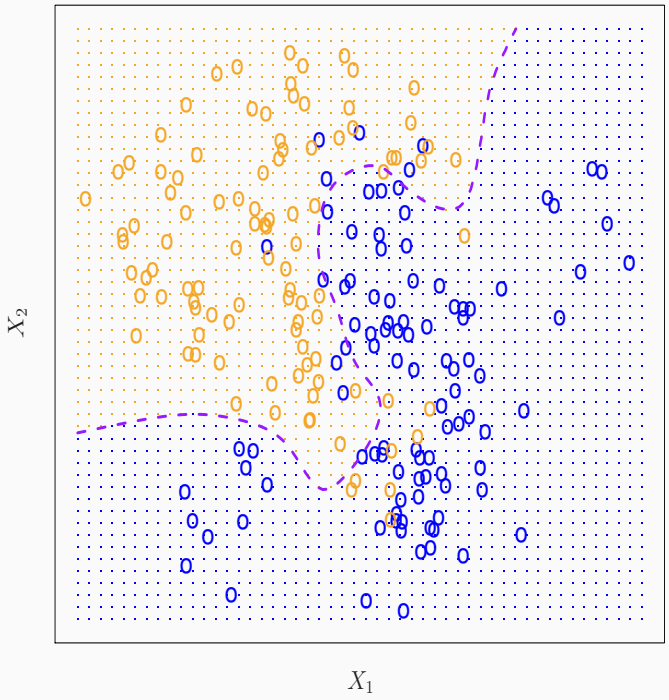

Figure 3: A simulated data set consisting of 100 observations in each of two groups, indicated in blue and in orange. The purple dashed line represents the Bayes decision boundary.

Bayes Classifier

Suppose $P(Y \mid X)$ is known. Then given an input x_0 we predict the categorical response y_0 as:

$$\hat{y}_0 = \underset{y}{\operatorname{argmax}}\, P(Y = y \mid X = x)$$

This Bayes classifier minimizes the expected 0-1 loss:

$$E[\frac{1}{m}\sum_{i=1}^{m} \mathbf{1}(\hat{y}_i \neq y_i)]$$

The minimum expected 0-1 loss is the Bayes error rate $1 - E(\operatorname{argmax}_y P(Y = y \mid X = x))$.

This is the best we can hope for - analogous to irreducible error in regression.

k-nearest neighbours

Bayes Classifier to k-NN

- In theory we would always like to predict qualitative responses using the Bayes classifier.
- But for real data, we do not know $P(Y \mid X)$, and so computing the Bayes classifier is impossible.
- It serves as an unattainable gold standard against which to compare other methods (also called 'Oracle').

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- In theory we would always like to predict qualitative responses using the Bayes classifier.
- But for real data, we do not know $P(Y \mid X)$, and so computing the Bayes classifier is impossible.
- It serves as an unattainable gold standard against which to compare other methods (also called 'Oracle').
- But we could estimate $P(Y \mid X)$ and then classify a given observation to the class with highest estimated probability.
- One such method is K-nearest neighbour.

k-nearest neighbor

- Given a positive integer K and a test observation x_0, the KNN classifier first identifies the K points in the training data that are closest to x_0, represented by N_0.
- It then estimates the conditional probability for class j as the fraction of points in $\mathbf{N}_0$ whose response values equal j:

$$Pr(Y = j \mid X = x_0) = \frac{1}{K} \sum_{i \in \mathbf{N}_0} I(y_i = j)$$

- Finally, KNN applies Bayes rule and classifies the test observation x_0 to the class with the largest probability.

KNN Classifier

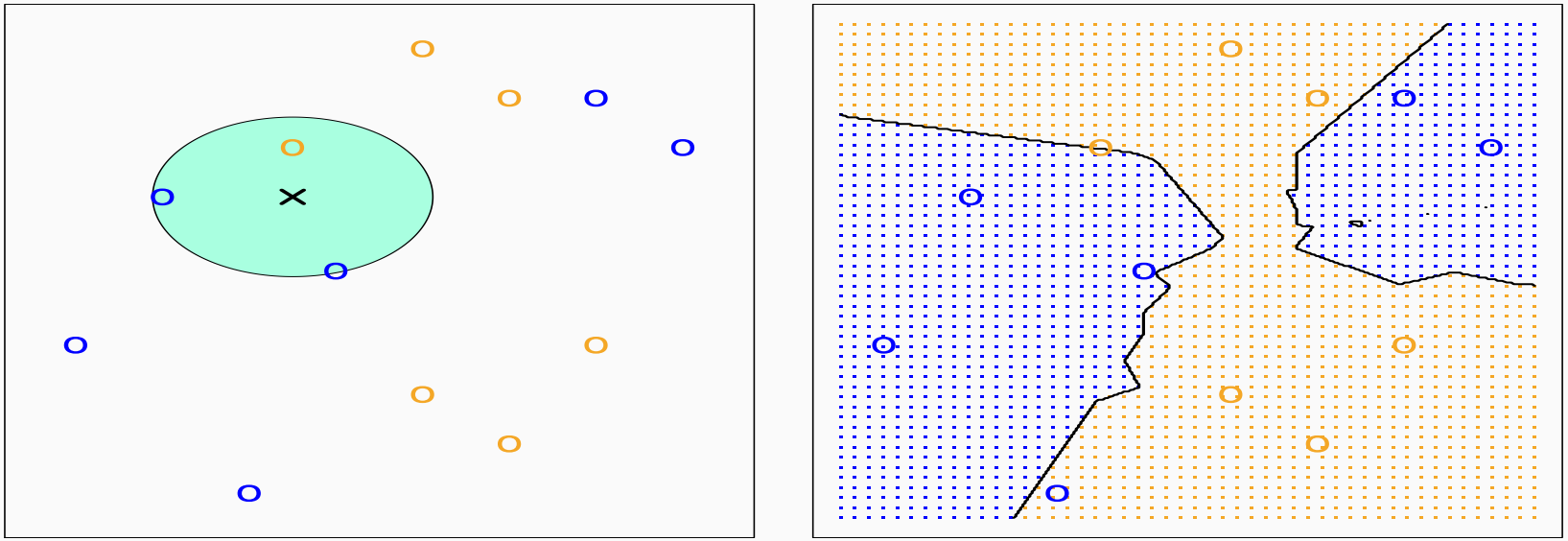

Figure 5: The KNN approach, using $K = 3$, is illustrated in a simple situation with six blue observations and six orange observations. Left: a test observation at which a predicted class label is desired is shown as a black cross. The three closest points to the test observation are identified, and it is predicted that the test observation belongs to the most commonly-occurring class, in this case blue. Right: The KNN decision boundary for this example is shown in black.

How good is KNN?

Despite the fact that it is a very simple approach, KNN can often produce classifiers that are surprisingly close to the optimal Bayes classifier.

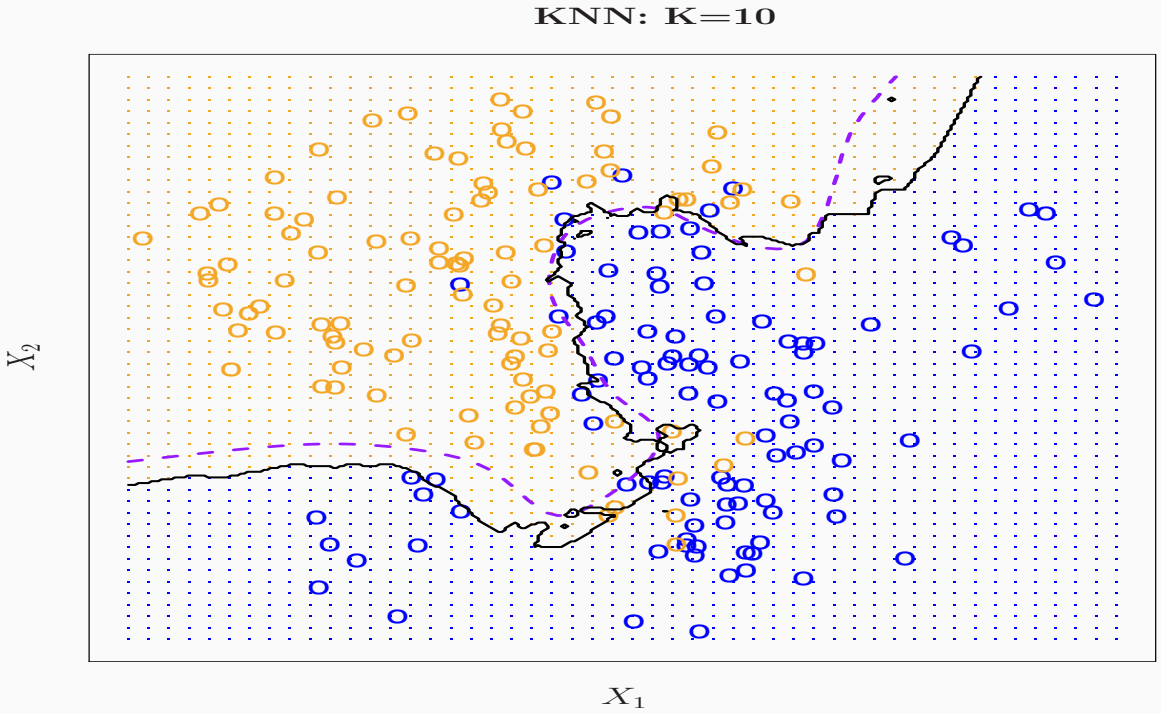

Figure 6: The black curve indicates the KNN decision boundary using K = 10. The Bayes decision boundary is shown as a purple dashed line. The KNN and Bayes decision boundaries are very similar

K acts as a smoothing parameter

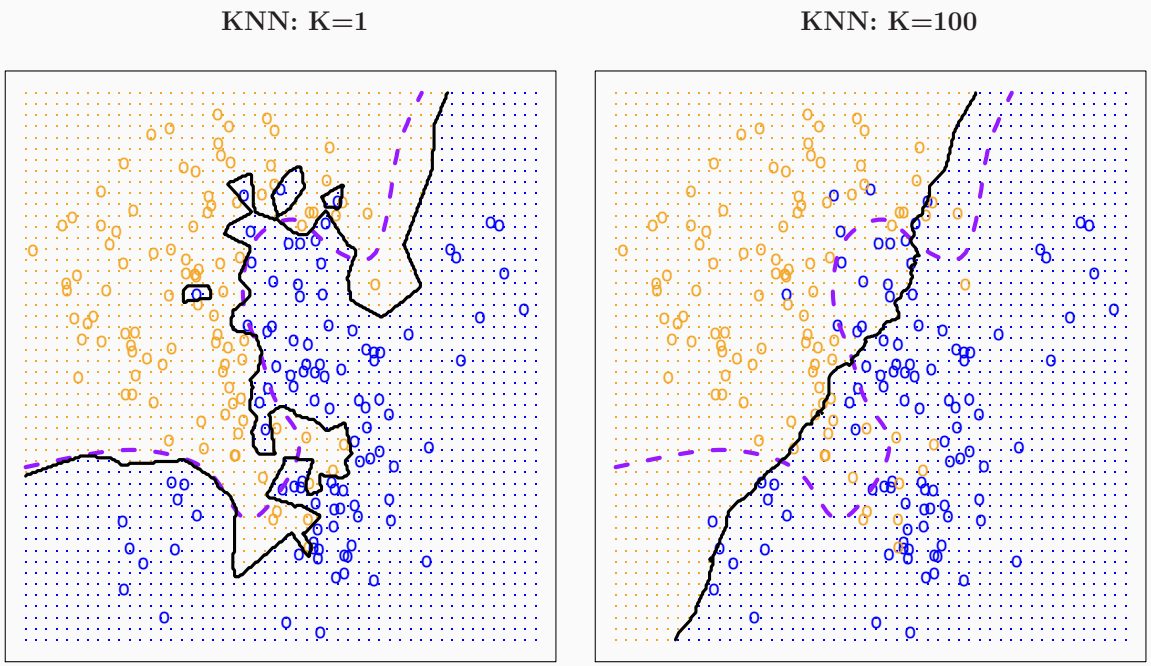

Figure 7: Comparison of the KNN decision boundaries (solid black curves) obtained using K = 1 and K = 100 on the same data. With $K = 1$, the decision boundary is overly flexible, while with $K = 100$ it is not sufficiently flexible. The Bayes decision boundary is shown as a purple dashed line.

Choice of K is important !!

$K = 1$, overly flexible decision boundary :low bias but very high variance.
As K grows, less flexible boundary (close to linear): low-variance but high-bias.

K acts as a smoothing parameter

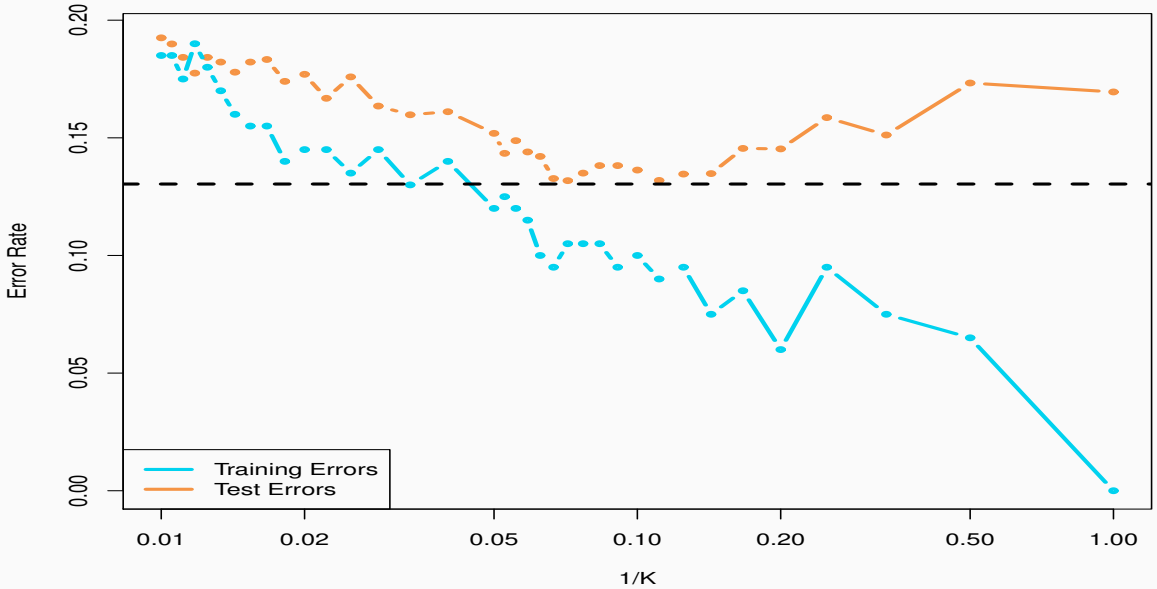

Figure 8: The KNN training error rate (blue, 200 observations) and test error rate (orange, 5,000 observations), as the level of flexibility (assessed using $1/K$) increases. The black dashed line indicates the Bayes error rate. The jumpiness of the curves is due to the small training data.

Choice of K: Hall P, Park B, Samworth R (2008). "Choice of neighbor order in nearest-neighbor classification". Annals of Statistics.

KDE vs. KNN

- Histogram methods partition the data space into distinct bins with width $h_i = (a_{i+1} - a_i)$ and count the number of observations, n_i.
- The estimate of density in each bin: $p_i = \frac{n_i}{nh_i}$,
- Often the bin-width is fixed: $h_i = h$.
- This h is our "smoothing parameter".

Basic idea

- Assume observations are drawn from some density $f(x)$ and consider a small region R containing x such that:

$$p = \int_R f(x)dx.$$

- The number of points k out of total n points that will lie inside this region R will follow a Binomial distribution with size n and probability p and a simple estimate is given by the mean $\hat{k} = n \times p$.
- If R is also very small so that $f(x)$ is practically constant over R, then we can also approximate $p \approx f(x) \times \text{size of } R = f(x) \times V$, implying

$$p = f(x) \times V = \frac{k}{n} \Rightarrow f(x) = \frac{k}{nV}.$$

KDE

- Histogram simply counts the number of observations inside R, KDE assigns them individual weight depending on their distance from the point x where you want $f(x)$.
- Kernel Density Estimate:

$$\hat{f}(x) = \frac{1}{2nh} \sum_{i=1}^{n} K(\frac{X_i - x}{h}).$$

  where $K(\cdot)$ is a Kernel function, $X_1, \ldots, X_n$ are observations.
- Choices of kernel function
  1. Uniform: $K(u) = 1/2I(u \leq 1)$: reduces to histogram.
  2. Gaussian: $K(u) = \phi(u)$ (standard normal: smoother)
- Same formula ($f(x) = k/nV$), with $k = \sum_i K(\frac{X_i - x}{h})$.

Nearest neighbour

- Think of nearest neighbour as a density estimation strategy.
- in KDE or histogram, we fix an interval or region (V) and calculate how many points lie within (K).
- In K-nearest neighbour, you fix the number of neighbours (K) centered on x, and let the region grow to a volume V^*. The region V^* will contain k out of n data points, as before, and we can get:

$$p(x) \approx \frac{k}{nV^*}$$

- In histogram of KDE, h acts as smoothing parameter, in k-NN, k acts as a smoothing parameter.

Classification is basically density estimation

- Suppose we have L classes $C_1, C_2, \ldots, C_L$: n_j observations from class C_j, i.e.

$$\sum_{j=1}^{L} n_j = n, \quad \text{and} \quad f(x) \approx \frac{k}{nV}.$$

- If we restrict ourselves to a fixed class, say C_j, then

$$f(x \mid C_j) \approx \frac{k_j}{n_j V}.$$

- Proportion of class j is $P(C_j) = n_j / n$.
- Now, we can apply Bayes' theorem:

$$P(C_j \mid x) = \frac{f(x \mid C_j) \times P(C_j)}{f(x)} = \frac{k_j / n_j V \times n_j / n}{\frac{k}{nV}} = \frac{k_j}{k}.$$

- Simpler words: how many points of class j out of the K nearest neighbours? Decide by majority.

Classification: Towards logistic regression

Strategy: Estimate $P(Y \mid X)$

- If we have a good estimate for the conditional probability $\hat{P}(Y \mid X)$, we can use it for the classifier.

$$\hat{y}_0 = \operatorname*{argmax}_j \hat{P}(Y = j \mid X = x_0)$$

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- Suppose Y is a binary variable. Could we use a linear model?

$$P(Y = 1 \mid X) = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p + \epsilon$$

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- No.
- This would allow probabilities < 0 and > 1. Difficult to extend to more than 2 categories.

History

Fisher proposed linear discriminant analysis in 1936.

In the 1940s, various authors put forth an alternative approach, logistic regression.

In the early 1970s, Nelder and Wedderburn coined the term *generalized linear models* for an entire class of statistical learning methods that include both linear and logistic regression as special cases.

Logistic Regression

- We model the probabilities as:

$$P(Y = 1 \mid X) = \frac{e^{\beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p}}$$

$$P(Y = 0 \mid X) = \frac{1}{1 + e^{\beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p}}$$

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$$P(Y = 0 \mid X) = \frac{1}{1 + e^{\beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p}}$$

- This is the same as using a linear model for the log odds:

$$\log \left[\frac{P(Y = 1 \mid X)}{P(Y = 0 \mid X)} \right] = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p$$

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- The left hand side is called log-odds or logit.

Different choices

- The general idea is that we need a function that maps the real line $\mathbb{R}$ to the unit interval.
- An excellent candidate is cumulative distribution functions of random variables $F(x) = P(X \leq x)$. CDF's are, by definition, right-continuous, non-decreasing, and will always satisfy $\lim_{x \to \inf} F(x) = 0$ and $\lim_{x \to \inf} F(x) = 1$.
- Some examples of CDF's we can use:

$$\begin{aligned} \text{Uniform} \quad & F(x) = x\mathbb{I}(0 \leq x \leq 1) \\ \text{Logistic} \quad & F(x) = \text{logistic}(x) = \frac{1}{1 + e^{-x}} \\ \text{Gaussian} \quad & F(x) = \Phi(x) \text{ (no closed form)} \end{aligned}$$

Interpreting β's

- In a linear regression model, β_1 gives the average change in Y associated with a one-unit increase in X.
- In contrast, in a logistic regression model, increasing X by one unit changes the log odds by a factor of β_1, or equivalently it multiplies the odds by e^{β_1}.

$$\log\left[\frac{P(Y=1 \mid X)}{P(Y=0 \mid X)}\right] = \beta_0 + \beta_1 X$$

- Because the relationship between $p(X)$ and X in is not a straight line, β_1 does not correspond to the change in $p(X)$ associated with a one-unit increase in X.

Fitting Logistic Regression

- The training data is a list of paired observations $(y_1, \mathbf{x}_1), \ldots, (y_n, \mathbf{x}_n)$. Need to estimate $\beta_0, \beta_1, \ldots, \beta_p$.
- In the linear model:

$$\log\left[\frac{P(Y=1 \mid X)}{P(Y=0 \mid X)}\right] = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p$$

  we don't observe the left hand side.
- We cannot use a least squares fit !
- We can use **maximum likelihood**.

Fitting

- Consider a simple model with one predictor:

$$\log\left[\frac{P(Y=1 \mid X)}{P(Y=0 \mid X)}\right] = \beta_0 + \beta_1 X$$

- Solution: The likelihood is the probability of the training data, for a fixed set of coefficients β_0, β_1:

$$\begin{aligned}
&\prod_{i=1}^{n} P(Y = y_i \mid X = x_i) \\
&= \underbrace{\prod_{i:y_i=1} \frac{e^{\beta_0+\beta_1 x_{i1}}}{1+e^{\beta_0+\beta_1 x_{i1}}}}_{\text{Probability of responses} = 1} \quad \underbrace{\prod_{i:y_i=0} \frac{1}{1+e^{\beta_0+\beta_1 x_{i1}}}}_{\text{Probability of responses} = 0}
\end{aligned}$$

- Choose estimates $\hat{\beta}_0, \hat{\beta}_1$ which maximize the likelihood.
- Solved with numerical methods (e.g. Newton's algorithm)

Issues with Logistic Regression

- The coefficients become unstable when there is collinearity. Furthermore, this affects the convergence of the fitting algorithm.
- When the classes are linearly separated (you can pass a hyperplane between them), the coefficients become unstable.
- This is always the case when $p \geq n - 1$.

A case study

https://dattahub.github.io/data-analytics/challenger_casestudy_ninja.html

Show R Examples

http://dattahub.github.io/data-analytics/logistic_demo.html

Deviance and Residuals

Example

```
> options(contrasts = c("contr.treatment", "contr.poly"))
> ldose <- rep(0:5, 2)
> numdead <- c(1, 4, 9, 13, 18, 20, 0, 2, 6, 10, 12, 16)
> sex <- factor(rep(c("M", "F"), c(6, 6)))
> SF <- cbind(numdead, numalive = 20 - numdead)
> budworm.lg <- glm(SF ~ sex*ldose, family = binomial)
> summary(budworm.lg, cor = F)
```

Example

```
Call:
glm(formula = SF ~ sex * ldose, family = binomial)

Deviance Residuals:
     Min        1Q    Median        3Q       Max
-1.39849  -0.32094  -0.07592   0.38220   1.10375

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -2.9935     0.5527  -5.416 6.09e-08 ***
sexM          0.1750     0.7783   0.225    0.822
ldose         0.9060     0.1671   5.422 5.89e-08 ***
sexM:ldose    0.3529     0.2700   1.307    0.191
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 124.8756  on 11  degrees of freedom
Residual deviance:   4.9937  on  8  degrees of freedom
AIC: 43.104

Number of Fisher Scoring iterations: 4
```

Standard errors

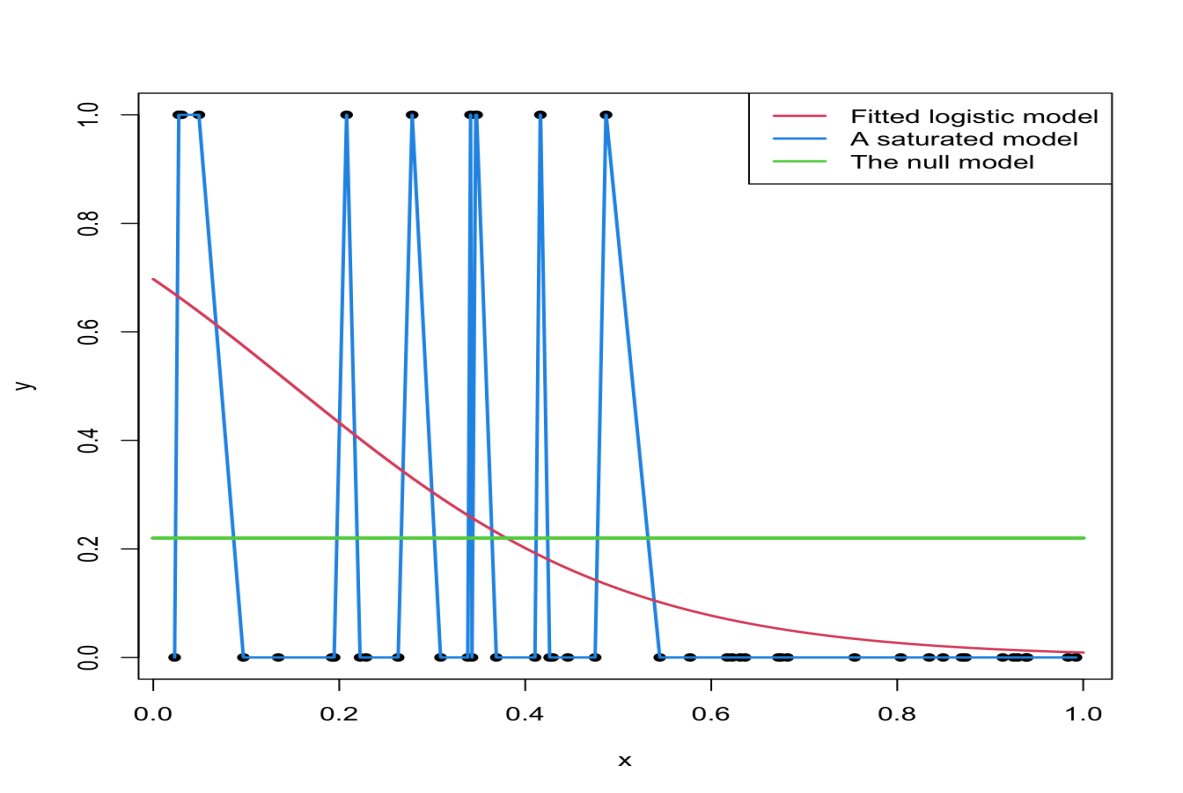
- We can estimate the Standard Error of each coefficient.
- The z-statistic is the equivalent of the t-statistic in linear regression:

$$z_j = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}$$

- The p-values are test of the null hypothesis $\beta_j = 0$ (Wald's test).
- Other possible hypothesis tests: likelihood ratio test (chi-square distribution) is useful for testing whether groups of variables have coefficients equal to 0.

Goodness of fit: Deviance

- Deviance measures the deviance of the fitted logistic model with respect to a 'perfect model' for $P(Y = 1|X)$.
- 'Perfect model': Saturated model $\hat{P}(Y = 1|X) = \hat{y}_i$, $i = 1, \ldots, n$.



Deviance i

- More precisely, the deviance is defined as the difference of likelihoods between the fitted model and the saturated model:

$$D = -2\text{loglik}(\hat{\beta}) + 2\text{loglik}(\text{saturated model})$$

- It can be shown that the $\text{loglik}(\text{saturated model}) = 1$, and hence the deviance can be simplified to:

$$D = -2\text{loglik}(\hat{\beta})$$

- the deviance is always larger or equal than zero, being zero only if the fit is perfect.
- Benchmark for evaluating the magnitude of the deviance is the null deviance: deviance of the worst model, the one fitted without any predictor, to the perfect model.

$$D_0 = -2\text{loglik}(\hat{\beta}_0), \text{ where } \hat{\beta_0} = \text{logit}(m/n), m = \sum_{i=1}^{n} y_i$$

Deviance ii

- The null deviance serves for comparing how much the model has improved by adding the predictors.
- This can be done by means of the R^2 statistic, which is a generalization of the determination coefficient in multiple linear regression:

$$R^2 = 1 - \frac{D}{D_0} = 1 - \frac{\text{deviance of fitted model to saturated}}{\text{deviance of null model to saturated}}.$$

  1. It is a quantity between 0 and 1.
  2. If the fit is perfect, then $D = 0$ and $R^2 = 1$, but if the predictors have no information about Y, then $D = D_0$, and $R^2 = 0$.
- But it is not the %-age of variance explained or related to correlation for logistic, like we interpret for linear regression.

Supplementary R demo

1. **R codes for deviance**: https://dattahub.github.io/data-analytics/deviance_explained.html

Extending Logistic Regression to more than 2 categories

- Multinomial Logistic Regression:
- Suppose Y takes values in $\{1, 2, \ldots, K\}$, then we use a linear model for the log-odds against a baseline category (e.g. category 1).

$$\log\left[\frac{P(Y=2 \mid X)}{P(Y=1 \mid X)}\right] = \beta_{0,2} + \beta_{1,2}X_1$$

$$\ldots$$

$$\log\left[\frac{P(Y=K \mid X)}{P(Y=1 \mid X)}\right] = \beta_{0,K} + \beta_{1,K}X_1$$

Supplementary R demo

1. **Example of a multi-class logistic regression**: https://dattahub.github.io/data-analytics/logistic_other.html

Linear Discriminant Analysis

Linear Discriminant Analysis

- In logistic regression, we estimate the conditional class probabilities $P(Y \mid X = k)$ directly.
- LDA is somewhat indirect approach, here we model probability of X given Y and use Bayes' theorem to calculate probability of Y given X.
- LDA has more assumptions: when X is assumed to be Normal, LDA behaves like logistic.

Linear Discriminant Analysis

Strategy: Instead of estimating $P(Y \mid X)$, we will estimate:

1. $P(X \mid Y)$: Given the response what is the distribution of the inputs.
2. $P(Y)$: How likely each category is?

Now we can use Bayes' Rule:

$$\begin{aligned}\hat{P}(Y = k \mid X = x) &= \frac{\hat{P}(X = x \mid Y = k)\hat{P}(Y = k)}{\hat{P}(X = x)} \\ &= \frac{\hat{P}(X = x \mid Y = k)\hat{P}(Y = k)}{\sum_j \hat{P}(X = x \mid Y = k)\hat{P}(Y = k)}\end{aligned}$$

Linear Discriminant Analysis

Strategy: Instead of estimating $P(Y \mid X)$, we will estimate:

1. $\hat{P}(X \mid Y) = \hat{f}_k(x)$, where each $\hat{f}_k(x)$ is a multivariate Normal density.
2. $\hat{P}(Y = k) = \hat{\pi}_k =$ the fraction of training samples of class k.

$$X_{p\times 1} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \text{ Multivariate Normal with } \mathbb{E}(X) = \boldsymbol{\mu}, \mathbf{V}(X) = \boldsymbol{\Sigma}$$

$$\text{if } f(x) = \frac{1}{(2\pi)^{p/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$$

Multivariate Normal

If $Var(X_1) = Var(X_2)$ and $Cor(X_1, X_2) = 0$; this surface has a characteristic bell shape. However, the bell shape will be distorted if the predictors are correlated or have unequal variances. In this situation, the base of the bell will have an elliptical, rather than circular, shape.

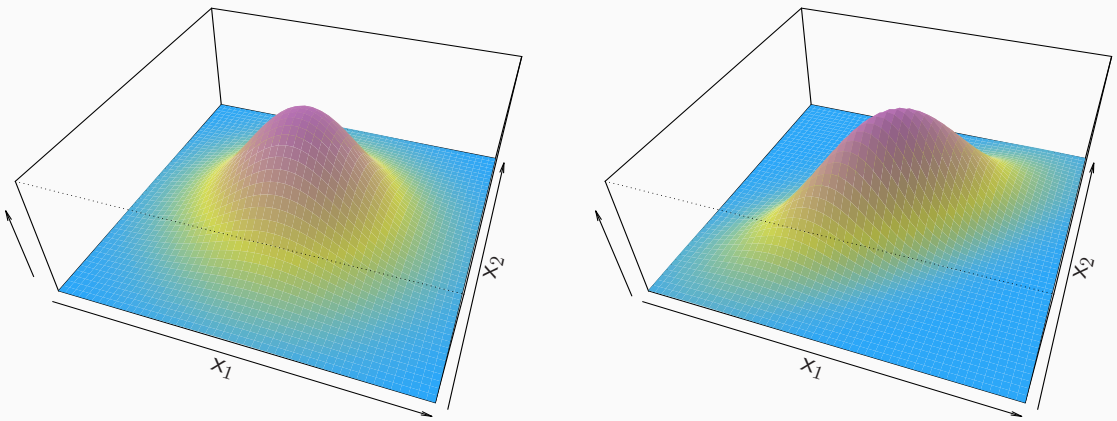


Figure 9: Two multivariate Gaussian density functions are shown, with p = 2. Left: The two predictors are uncorrelated. Right: The two variables have a correlation of 0.7.

Linear Discriminant Analysis

Strategy: Instead of estimating $P(Y \mid X)$, we will estimate:

1. $\hat{P}(X \mid Y) = \hat{f}_k(x)$, where each $\hat{f}_k(x)$ is a multivariate Normal density.
2. $\hat{P}(Y = k) = \hat{\pi}_k =$ the fraction of training samples of class k.

Questions:

1. How do we estimate the parameters of the multivariate Normal distribution?
2. What do LDA predictions look like?

Univariate Normal

Left: Two one-dimensional normal density functions are shown. The dashed vertical line represents the Bayes decision boundary. Right: 20 observations were drawn from each of the two classes, and are shown as histograms.

The Bayes decision boundary is again shown as a dashed vertical line. The solid vertical line represents the LDA decision boundary estimated from the training data.

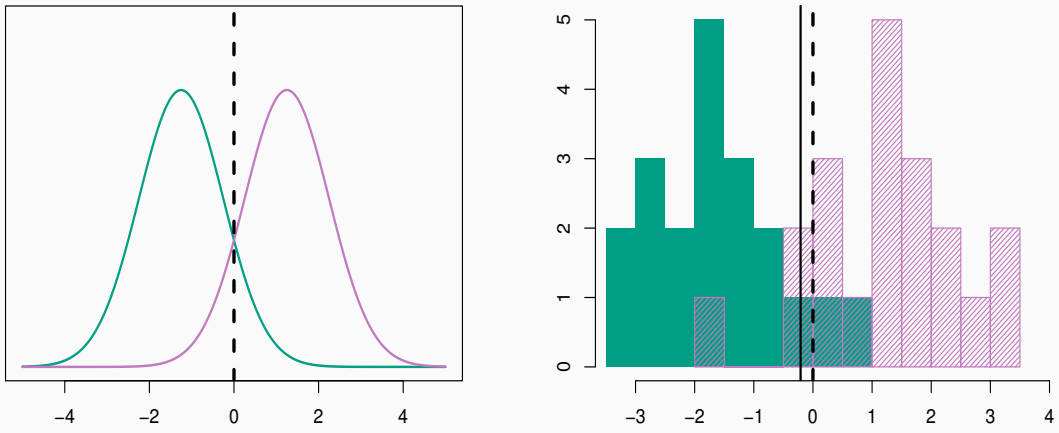


Figure 10: Two univariate Gaussian density functions are shown, with $p = 1$.

Multivariate Normal

If $Var(X_1) = Var(X_2)$ and $Cor(X_1, X_2) = 0$; this surface has a characteristic bell shape. However, the bell shape will be distorted if the predictors are correlated or have unequal variances. In this situation, the base of the bell will have an elliptical, rather than circular, shape.

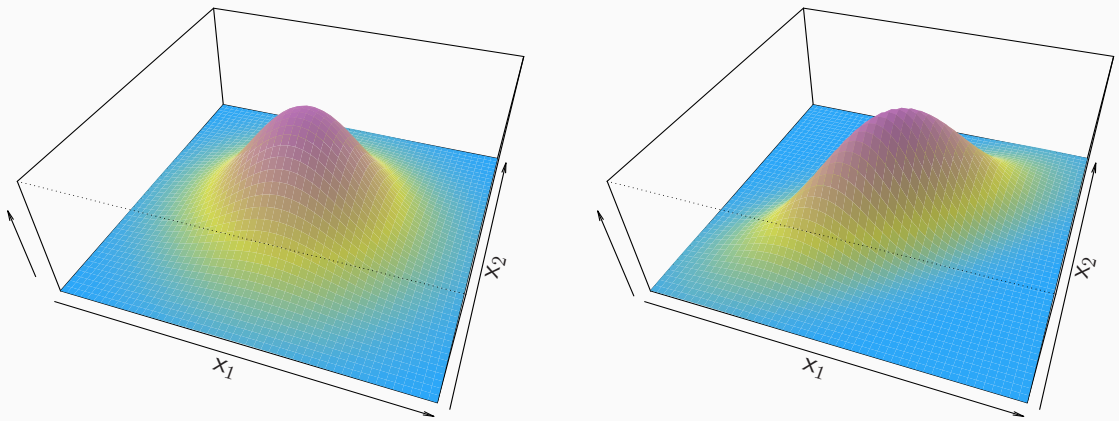


Figure 11: Two multivariate Gaussian density functions are shown, with p = 2. Left: The two predictors are uncorrelated. Right: The two variables have a correlation of 0.7.

LDA has linear decision boundaries

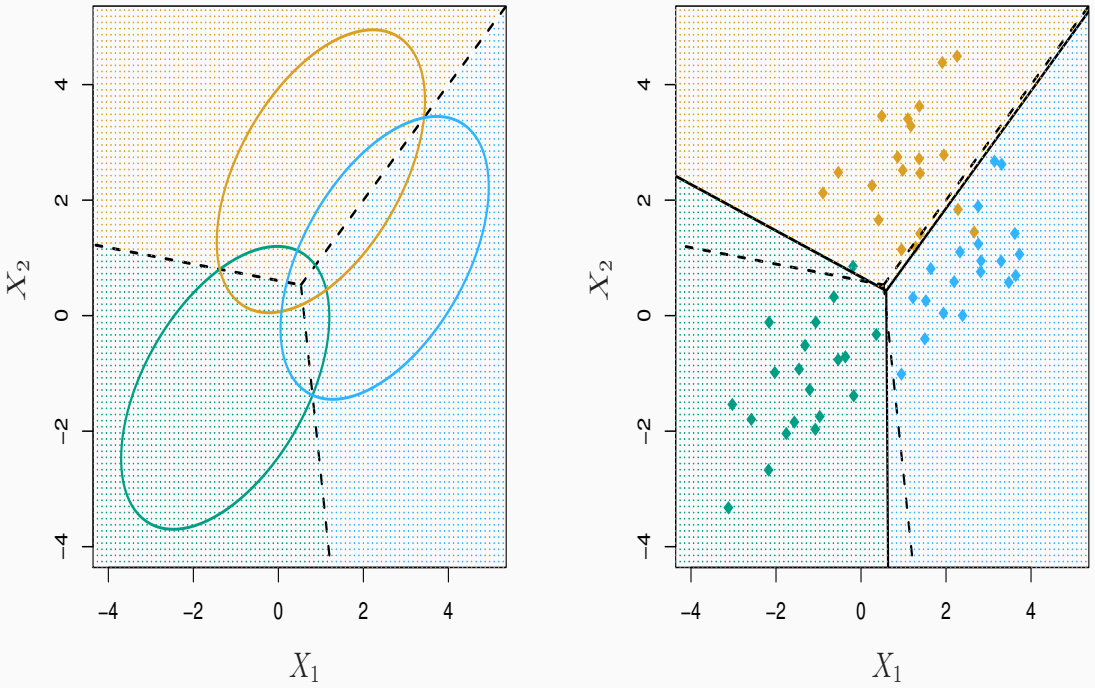


Figure 12: Left: Ellipses that contain 95% of the probability for each of the three classes are shown. The dashed lines are the Bayes decision boundaries. Right: 20 observations were generated from each class, and the corresponding LDA decision boundaries are indicated using solid black lines. The Bayes decision boundaries are once again shown as dashed lines.

LDA for $k = 1$ i

- For $k = 1$, f_k is a univariate normal, let the mean and variance be μ_k and $\sigma_k^2 = \sigma^2, \forall k = 1, 2, \ldots, K$.

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} \exp\left(-\frac{1}{2\sigma_k^2}(x - \mu_k)^2\right)$$

- By Bayes rule, the probability of category k, given the input x is:

$$P(Y = k \mid X = x) = \frac{\textit{likelihood} \times \textit{prior}}{\textit{marginal}} = \frac{f_k(x)\pi_k}{P(X = x)} = \frac{f_k(x)\pi_k}{\sum_{l=1}^{K} \pi_l f_l(x)}$$

- Plugging in the formula for $f_k(x)$:

$$P(Y = k \mid X = x) = \frac{\pi_k \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_k)^2\right)}{\sum_{l=1}^{K} \pi_l \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu_l)^2\right)} \quad (4)$$

LDA for $k = 1$ ii

- The Bayes classifier involves assigning an observation $X = x$ to the class for which (4) is largest.
- Taking the log and rearranging the terms, it is not hard to show that this is equivalent to assigning the observation to the class for which:

$$\delta_k(x) = x\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

is the largest.

LDA for $k \geq 2$

Suppose that:

- We know $P(Y = k) = \pi_k$ exactly.
- $P(X \mid Y = k)$ is multivariate Normal with density:

$$f_k(x) = \frac{1}{(2\pi)^{p/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}_k)\right)$$

- $\boldsymbol{\mu}_k$: Mean of category k.
- $\boldsymbol{\Sigma}$: Covariance matrix (common to all categories k)
- Then, what is the Bayes classifier?

LDA has linear decision boundaries

By Bayes rule, the probability of category k, given the input x is:

$$P(Y = k \mid X = x) = \frac{\textit{likelihood} \times \textit{prior}}{\textit{marginal}} = \frac{f_k(x)\pi_k}{P(X = x)}$$

The denominator does not depend on the response k, so we can write it as a constant

$$P(Y = k \mid X = x) = C \times f_k(x)\pi_k$$

Plugging in the formula for $f_k(x)$:

$$P(Y = k \mid X = x) = \frac{C\pi_k}{(2\pi)^{p/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}_k)\right)$$

LDA has linear decision boundaries

$$P(Y = k \mid X = x) = \frac{C\pi_k}{(2\pi)^{p/2}|\boldsymbol{\Sigma}|^{1/2}} e^{\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_k)^T\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}_k)\right)}$$

Now, let us absorb everything that does not depend on k into a constant C':

$$P(Y = k \mid X = x) = C'\pi_k \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_k)^T\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}_k)\right)$$

and take the logarithm of both sides:

$$\log P(Y = k \mid X = x) = \log C' + \log \pi_k - \frac{1}{2}(\mathbf{x}-\boldsymbol{\mu}_k)^T\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu}_k)$$

This constant is the same for every category, k.
So we want to find the maximum of this over k.

LDA has linear decision boundaries

- Goal, maximize the following over k:

$$\log \pi_k - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}_k)$$

LDA has linear decision boundaries

- Goal, maximize the following over k:

$$\log \pi_k - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}_k)$$
$$\log \pi_k - \frac{1}{2}[\mathbf{x}^T \boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\mu}_k^T \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k] + \mathbf{x}^T \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k$$

LDA has linear decision boundaries

- Goal, maximize the following over k:

$$\log \pi_k - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}_k)$$

$$\log \pi_k - \frac{1}{2}[\mathbf{x}^T \boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\mu}_k^T \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k] + \mathbf{x}^T \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k$$

$$C'' + \log \pi_k - \frac{1}{2}\boldsymbol{\mu}_k^T \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k + \mathbf{x}^T \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k$$

LDA has linear decision boundaries

- Goal, maximize the following over k:

$$\log \pi_k - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}_k)$$
$$\log \pi_k - \frac{1}{2}[\mathbf{x}^T \boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\mu}_k^T \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k] + \mathbf{x}^T \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k$$
$$C'' + \log \pi_k - \frac{1}{2}\boldsymbol{\mu}_k^T \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k + \mathbf{x}^T \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k$$

- We can define the objective function as:

$$\delta_k(x) = \log \pi_k - \frac{1}{2}\boldsymbol{\mu}_k^T \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k + \mathbf{x}^T \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k$$

LDA has linear decision boundaries

- Goal, maximize the following over k:

$$\log \pi_k - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}_k)$$
$$\log \pi_k - \frac{1}{2}[\mathbf{x}^T \boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\mu}_k^T \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k] + \mathbf{x}^T \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k$$
$$C'' + \log \pi_k - \frac{1}{2}\boldsymbol{\mu}_k^T \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k + \mathbf{x}^T \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k$$

- We can define the objective function as:

$$\delta_k(x) = \log \pi_k - \frac{1}{2}\boldsymbol{\mu}_k^T \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k + \mathbf{x}^T \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_k$$

- At an input x, we predict the response with the highest $\delta_k(x)$.

LDA has linear decision boundaries

- What is the decision boundary? It is the set of points in which 2 classes are equally probable:

$$\delta_k(x) = \delta_l(x)$$

$$\log \pi_l - \frac{1}{2}\mu_l^T \Sigma^{-1} \mu_l + \mathbf{x}^T \Sigma^{-1} \mu_l = \log \pi_k - \frac{1}{2}\mu_k^T \Sigma^{-1} \mu_k + \mathbf{x}^T \Sigma^{-1} \mu_k$$

- This is a linear equation in x.

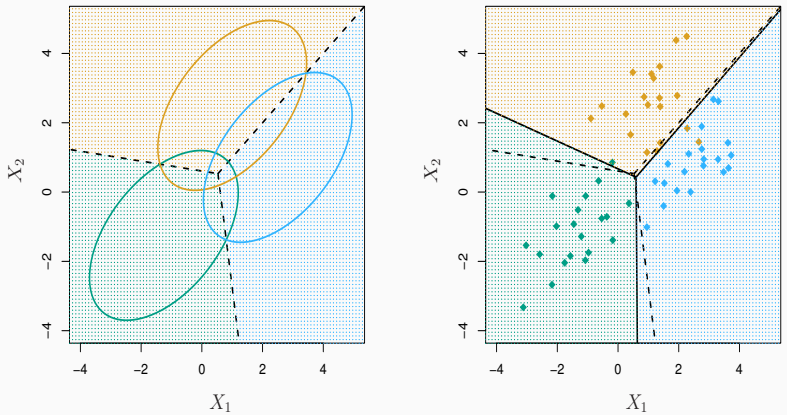


Figure 13: LDA decision boundaries are indicated using solid black lines. The Bayes decision boundaries are shown as dashed lines.

Estimating π_k

$$\hat{\pi}_k = \frac{\#i : y_i = k}{n}$$

In words, the fraction of training samples of class k.

Estimating the parameters of f_k

Estimate the mean vector μ_k for each class:

$$\hat{\mu}_k = \frac{1}{\#i : y_i = k} \sum_{i:y_i=k} x_i$$

Estimate the common covariance matrix $\mathbf{\Sigma}$ from data:

1. One predictor $p = 1$:

$$\hat{\sigma}^2 = \frac{1}{n-K} \sum_{k=1}^{K} \sum_{i:y_i=k} (x_i - \hat{\mu}_k)^2$$

2. Many predictors: Use an unbiased estimate of its covariance matrix $\mathbf{\Sigma}$ from the deviances $(x_1 - \hat{\mu}_{y_1}), (x_2 - \hat{\mu}_{y_2}), \ldots, (x_n - \hat{\mu}_{y_n})$.

LDA prediction

- Once we have the estimates $\hat{\pi}_k, \hat{\boldsymbol{\mu}}_k, \hat{\Sigma}$, we can predict using the estimated $\hat{\delta}_k$.

LDA prediction

- Once we have the estimates $\hat{\pi}_k, \hat{\boldsymbol{\mu}}_k, \hat{\Sigma}$, we can predict using the estimated $\hat{\delta}_k$.
- At an input x, we predict the response with the highest $\delta_k(x)$.

$$\hat{\delta}_k(x) = \log \hat{\pi}_k - \frac{1}{2}\hat{\boldsymbol{\mu}}_k^T \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}_k + \mathbf{x}^T \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}_k$$

LDA prediction

- Once we have the estimates $\hat{\pi}_k, \hat{\boldsymbol{\mu}}_k, \hat{\Sigma}$, we can predict using the estimated $\hat{\delta}_k$.
- At an input x, we predict the response with the highest $\delta_k(x)$.

$$\hat{\delta}_k(x) = \log \hat{\pi}_k - \frac{1}{2}\hat{\boldsymbol{\mu}}_k^T \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}_k + \mathbf{x}^T \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}_k$$

- The decision boundaries are defined by: $\hat{\delta}_k(x) = \hat{\delta}_l(x)$.

Quadratic Discriminant Analysis

Quadratic Discriminant Analysis

The assumption that the inputs of every class have the same covariance $\mathbf{\Sigma}$ can be quite restrictive:

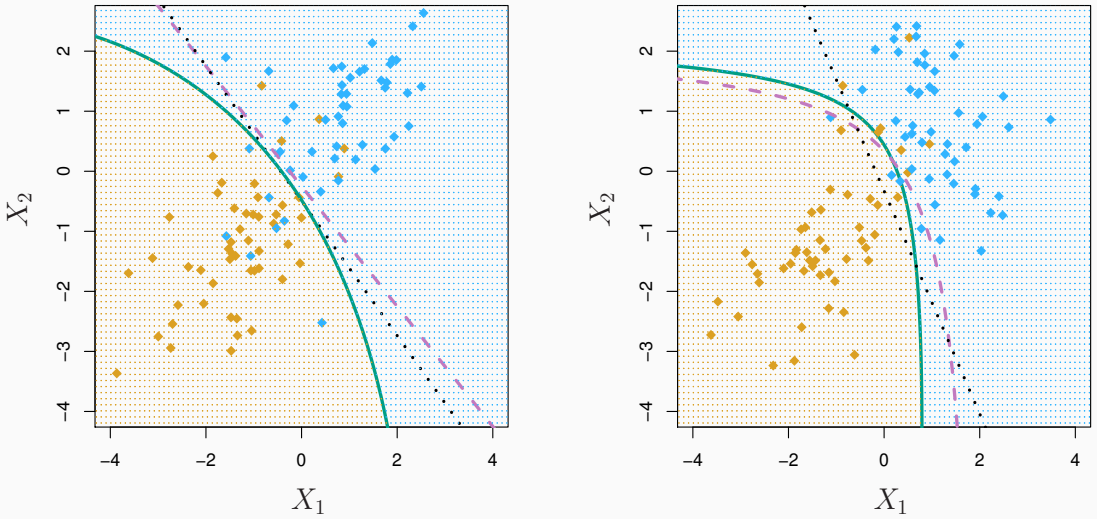


Figure 14: Left: $\mathbf{\Sigma}_1 = \mathbf{\Sigma}_2$, Right: $\mathbf{\Sigma}_1 \neq \mathbf{\Sigma}_2$

Boundaries for Bayes (dashed), LDA (dotted), and QDA (solid).

QDA

In quadratic discriminant analysis we estimate a mean $\hat{\mu}_k$ and a covariance matrix $\mathbf{\Sigma}_k$ for each class separately. Given an input, it is easy to derive an objective function:

$$\log \pi_k - \frac{1}{2}\mu_k^T \mathbf{\Sigma}_k^{-1}\mu_k + \mathbf{x}^T \mathbf{\Sigma}_k^{-1}\mu_k - \frac{1}{2}\mathbf{x}^T \mathbf{\Sigma}_k^{-1}\mathbf{x} - \frac{1}{2}\log|\Sigma_k|$$

We'll skip the details!

Evaluating a Classifier

the expected 0-1 loss:

$$\mathbb{E}[\frac{1}{m}\sum_{i=1}^{m}\mathbf{1}(\hat{y}_i \neq y_i)]$$

It is possible to make the wrong prediction for some classes more often than others.

The 0-1 loss doesn't tell you anything about this. A much more informative summary of the error is a confusion matrix:

		Predicted class		
		− or Null	+ or Non-null	Total
True class	− or Null	True Neg. (TN)	False Pos. (FP)	N
	+ or Non-null	False Neg. (FN)	True Pos. (TP)	P
	Total	N*	P*	

Example: Predicting default

- Used LDA to predict credit card default in a dataset of 10K people.
- Predicted "yes" if $P(\text{default} = \text{yes} \mid X) > 0.5$

		True default status		
		No	Yes	Total
Predicted default status	No	9,644	252	9,896
	Yes	23	81	104
	Total	9,667	333	10,000

- The error rate among people who do not default (false positive rate) is very low.
- However, the error rate among people who do default (false negative rate) is 76%.
- False negatives may be a bigger source of concern!
- One possible solution: Change the threshold.

Example: Predicting default

Changing the threshold to 0.2 makes it easier to classify to "yes".
Predicted "yes" if $P(\text{default} = \text{yes} \mid X) > 0.2$

		True default status		
		No	Yes	Total
Predicted default status	No	9,432	138	9,570
	Yes	235	195	430
	Total	9,667	333	10,000

Note that the rate of false positives became higher! That is the price to pay for fewer false negatives.

Let's visualize the dependence of the error on the threshold:

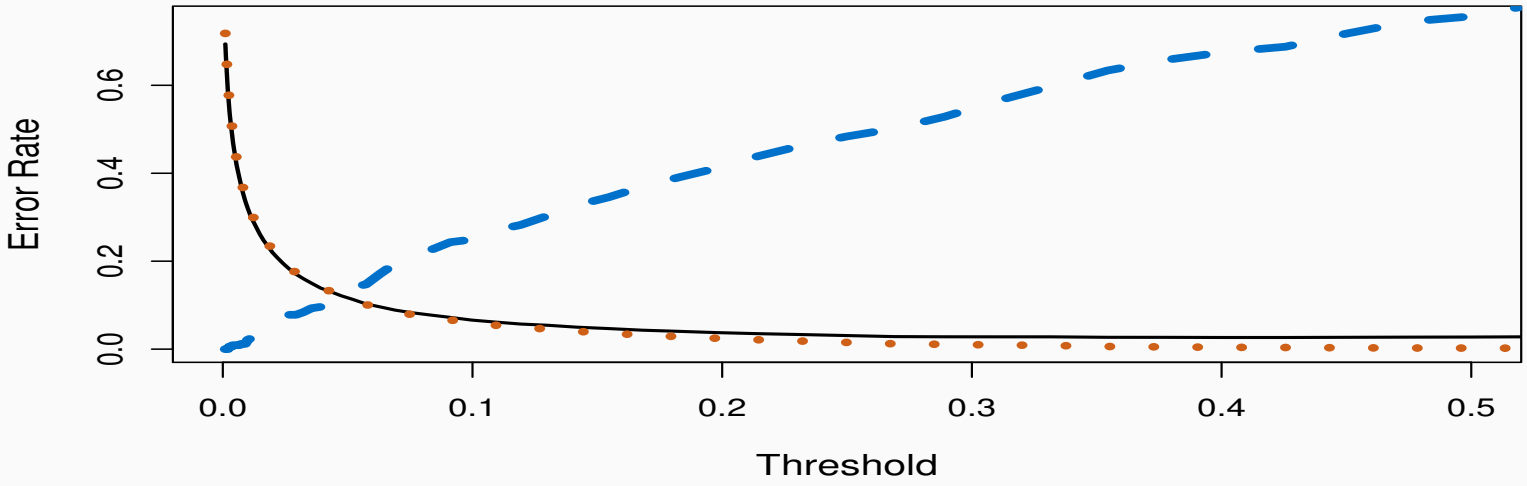


Figure 15: Error rates are shown as a function of the threshold value for the posterior probability that is used to perform the assignment. The **black solid line** displays the overall error rate. The blue dashed line represents the fraction of defaulting customers that are incorrectly classified, and the orange dotted line indicates the fraction of errors among the non-defaulting customers.

The Bayes Classifier

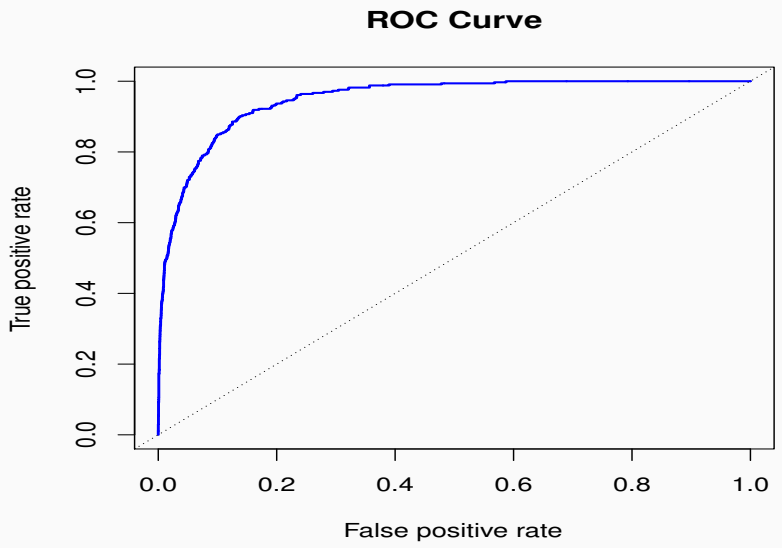


Figure 16: A ROC curve. The ideal ROC curve hugs the top left corner, indicating a high true positive rate and a low false positive rate. The dotted line represents the 'no information' classifier

- Displays the performance of the method for any choice of threshold.
- The area under the curve (AUC) measures the quality of the classifier:
- 0.5 is the AUC for a random classifier
- The closer AUC is to 1, the better.

Envoi

- Most of the regression methods we've studied aim to minimize the RSS, while classification methods aim to minimize the 0-1 loss.
- In classification, we often care about certain kinds of error more than others; i.e. the natural loss function is not the 0-1 loss.
- Changing the classification threshold to trade off between false positives and false negatives is an example of tuning a classifier to perform well with respect to a problem-specific objective.
- e.g. Find the threshold that brings the False negative rate below an acceptable level.

Show R Examples

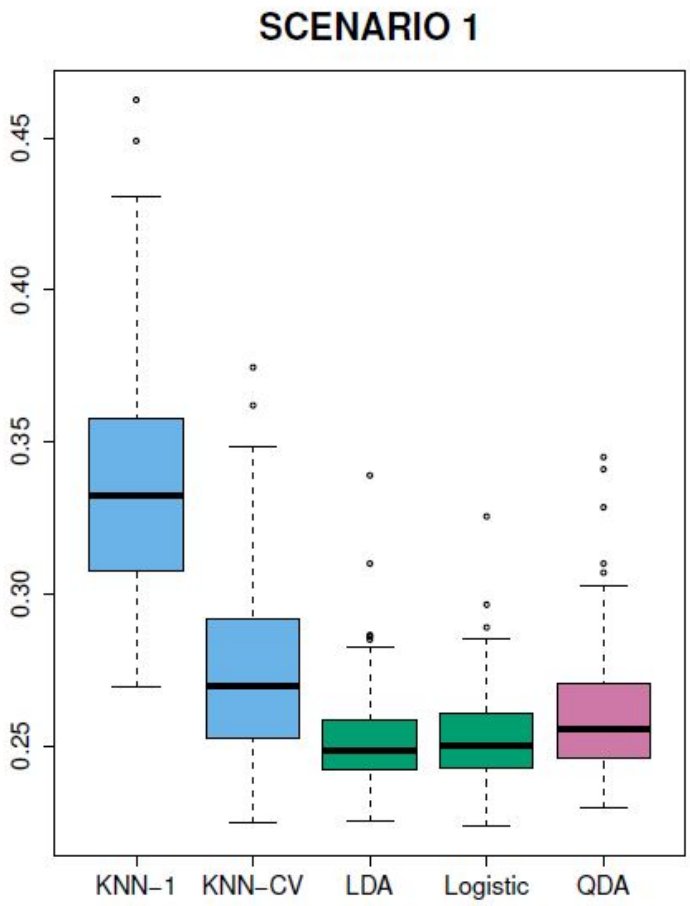
Compare different classification methods

Comparing

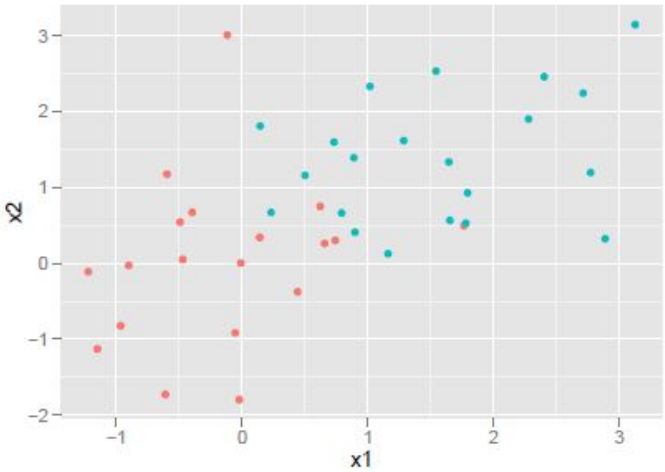
1. Simulate 100 random training sets from several different known distributions with 2 predictors and a binary response variable.
2. Compare the test error (0-1 loss) for the following methods:
   - 2.1 KNN-1
   - 2.2 KNN-CV (parameter K selected using cross-validation)
   - 2.3 Logistic regression
   - 2.4 Linear discriminant analysis (LDA)
   - 2.5 Quadratic discriminant analysis (QDA)

Check that the logistic regression boundary is linear!

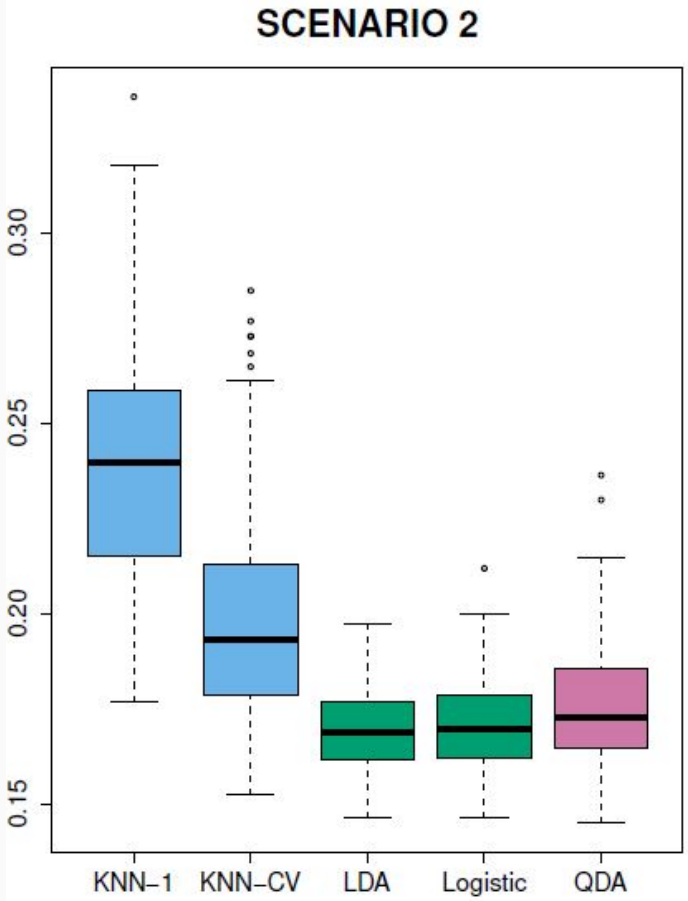
Scenario 1



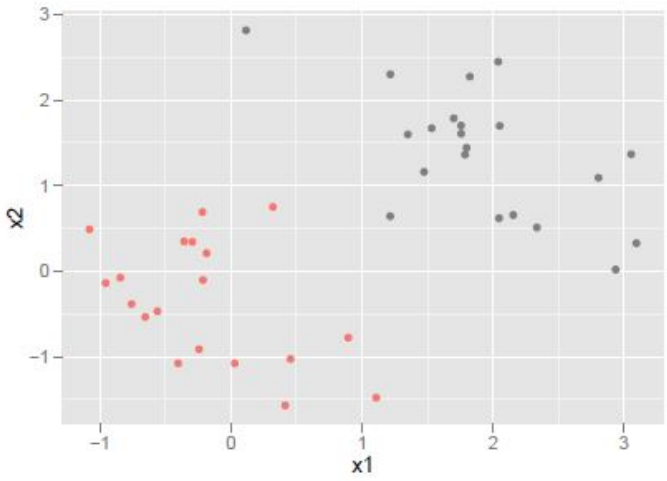
- In each class, (X_1, X_2) uncorrelated and bivariate normal, means differ between classes.



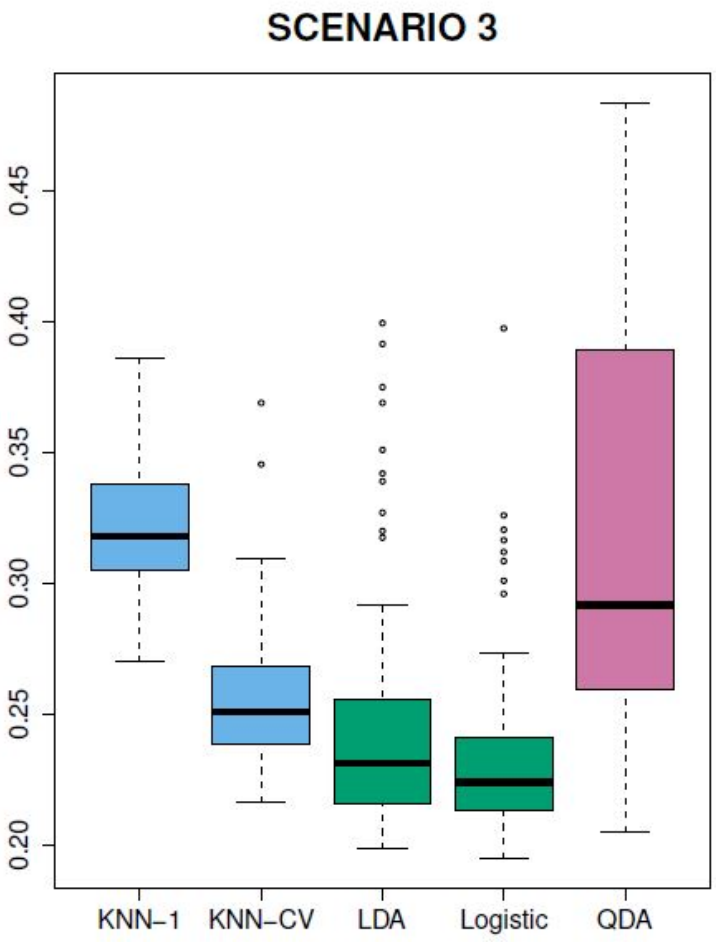
Scenario 2



- In each class, (X_1, X_2) is bivariate normal with correlation -0.5, means differ between classes.

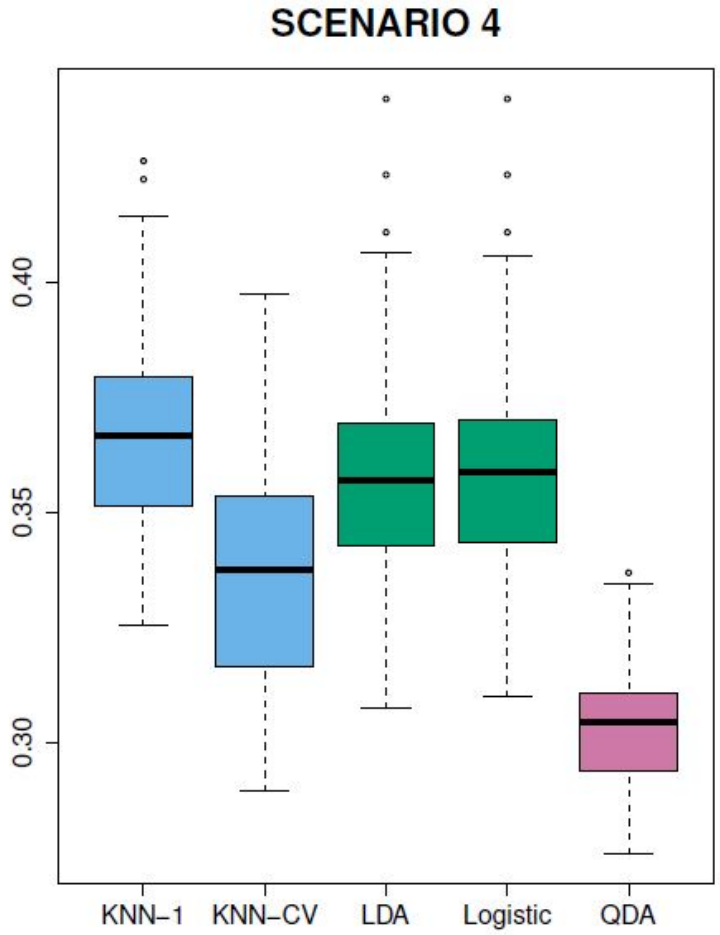


Scenario 3

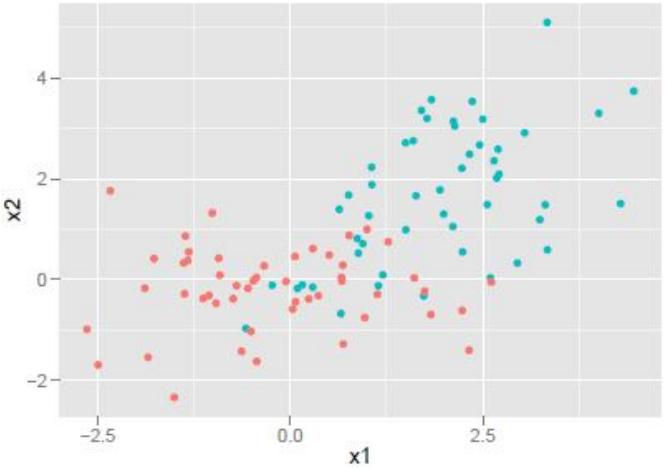


- In each class, (X_1, X_2) independent Student t random variables, means differ between classes.

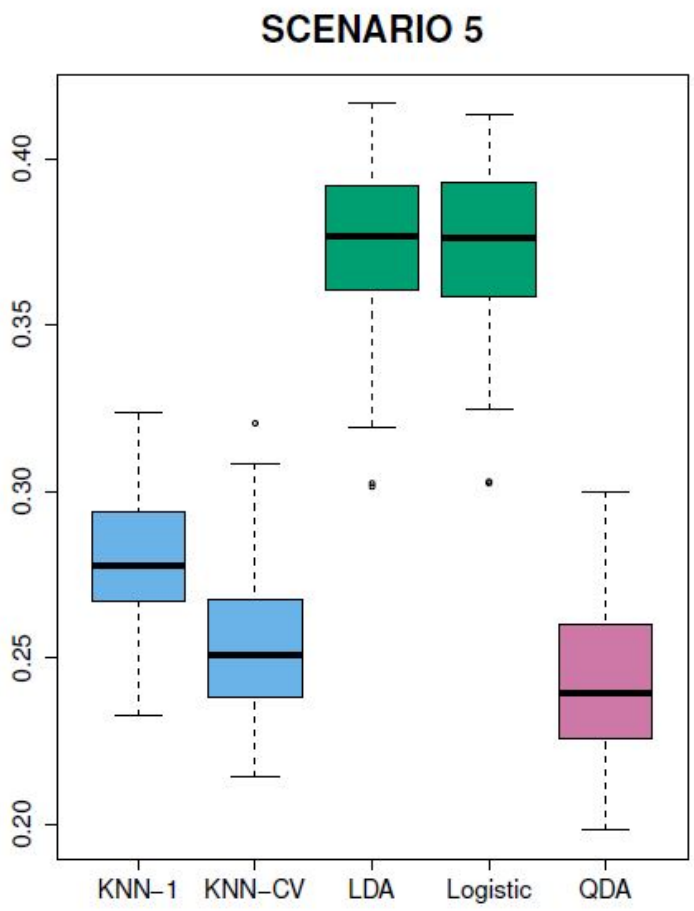
Scenario 4



- In each class, (X_1, X_2) bivariate normal, means differ between classes.
- First class has correlation 0.5, second class has correlation -0.5.



Scenario 5

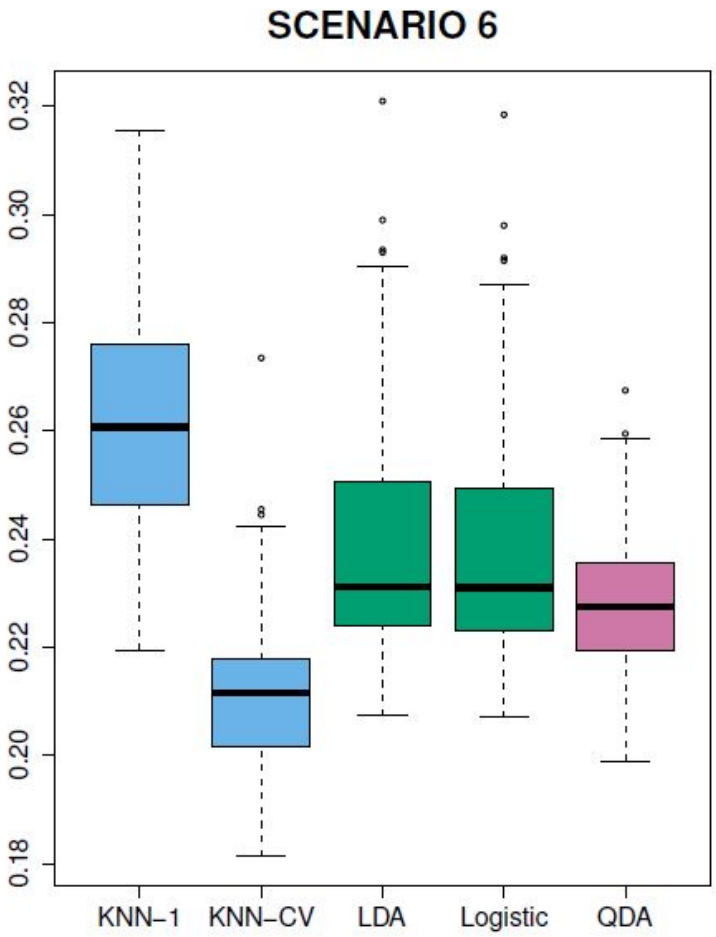


- X_1, X_2 uncorrelated, standard normal.
- Response Y was sampled from:

$$P(Y=1|X) = \frac{e^{\beta_0+\beta_1(X_1^2)+\beta_2(X_2^2)+\beta_3(X_1X_2)}}{1+e^{\beta_0+\beta_1(X_1^2)+\beta_2(X_2^2)+\beta_3(X_1X_2)}}$$

- The true decision boundary is quadratic.
- Note: We could also modify logistic regression to include quadratic terms, in the same way that we modified linear regression!

Scenario 6



- X_1, X_2 uncorrelated, standard normal.
- Response Y was sampled from:

$$P(Y = 1|X) = \frac{e^{f_{\text{nonlinear}}(X_1,X_2)}}{1 + e^{f_{\text{nonlinear}}(X_1,X_2)}}.$$

- The true decision boundary is very rough.

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- It's hard to choose a best method, but:
- When the true decision boundaries are linear, then the LDA and logistic regression approaches will tend to perform well.
- When the boundaries are moderately non-linear, QDA may give better results.
- Finally, for much more complicated decision boundaries, a non-parametric approach such as KNN can be superior. But the level of smoothness for a non-parametric approach **must be chosen carefully**.